

Part B — Work Breakdown Structure

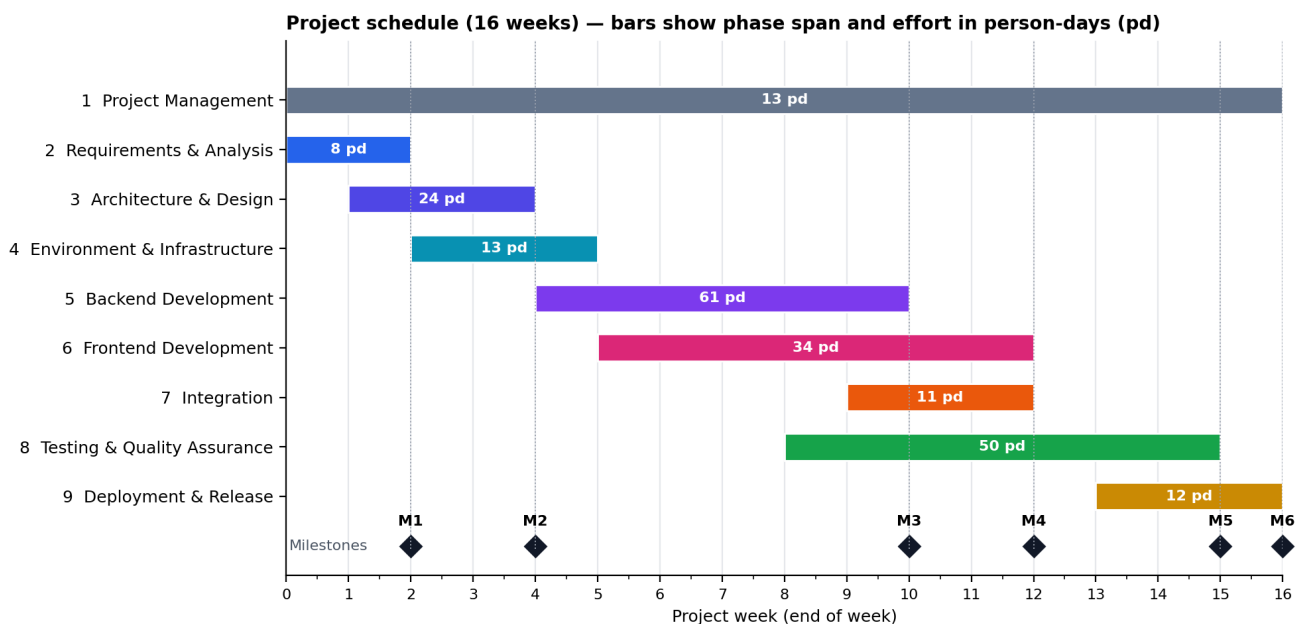
B.1 Overview and planning assumptions

- Delivery is planned over **16 weeks** for a team of five: project manager / business analyst, two backend developers, one frontend developer and one QA / DevOps engineer.
- Estimated effort is **226 person-days** plus a 15% contingency of 34 person-days, giving **260 person-days**. Effort is a first estimate and should be rescaled to your actual team.
- The WBS is organized by phase (level 1) and work packages (level 2). Each work package lists its deliverable, the requirements it realises, predecessors and effort.
- Phases overlap where dependencies allow: design starts before requirements sign-off completes, frontend starts once the API contract exists, and test preparation starts during build.

B.2 Phase summary

WBS	Phase	Weeks	Work packages	Effort (pd)	Share
1	Project Management	1-16	3	13	6%
2	Requirements & Analysis	1-2	4	8	4%
3	Architecture & Design	2-4	6	24	11%
4	Environment & Infrastructure	3-5	4	13	6%
5	Backend Development	5-10	11	61	27%
6	Frontend Development	6-12	8	34	15%
7	Integration	10-12	3	11	5%
8	Testing & Quality Assurance	9-15	8	50	22%
9	Deployment & Release	14-16	3	12	5%
	Total (before contingency)	1-16	50	226	100%
	Contingency (15%)			34	
	Planned total			260	

B.3 Schedule and milestones



Milestone	Description	Target week	Exit criteria
M1	SRS baseline signed off	2	SRS v1.0 approved; priorities and NFR targets confirmed (WBS 2.4)
M2	Architecture & design baseline	4	Architecture, API spec, schema and algorithm design reviewed
M3	Core services alpha	10	All seven services run end-to-end in the test environment
M4	Feature complete	12	All screens built; provider integrations working
M5	Test exit / UAT sign-off	15	No open critical or high defects; NFR benchmarks passed
M6	Go-live	16	Production release complete; hypercare started

B.4 Detailed work breakdown

pd = person-days. “Depends on” lists predecessor work packages; “5.x” means all packages in that phase.

WBS 1 — Project Management (weeks 1-16 · 13 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
1.1	Project planning & kickoff	Project plan, RACI, communication plan	—	—	3
1.2	Tracking, risk & status reporting	Weekly status reports, risk register	—	1.1	8
1.3	Final delivery & handover	Handover pack, lessons learned	—	9.3	2
	Phase total				13

WBS 2 — Requirements & Analysis (weeks 1-2 · 8 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
2.1	Finalize SRS from use-case flows	SRS v1.0 (this document)	FR-001-013	—	4
2.2	Confirm priorities & NFR targets	Approved priority and target list	NFR-001-011	2.1	1
2.3	Logical data model	ER diagram, data dictionary	FR-002, 003, 005	2.1	2
2.4	Requirements review & sign-off	Signed-off baseline + RTM	All	2.2, 2.3	1
	Phase total				8

WBS 3 — Architecture & Design (weeks 2-4 · 24 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
3.1	Architecture design & decisions	Architecture diagram, decision log	All	2.4	4
3.2	API specification	OpenAPI contract for all services	FR-002-013	3.1	4
3.3	Database schema design	DDL scripts, indexing plan	FR-002, 005, 006	2.3, 3.1	3
3.4	Settlement algorithm design	Algorithm spec, complexity analysis, test oracle	FR-001, 013, NFR-001, 005	2.4	4
3.5	UI/UX wireframes & prototype	Clickable prototype for 5 key screens	FR-002, 005, 007, 009	2.4	6
3.6	Security & access-control design	Threat model, RBAC matrix, key management	NFR-002, 006, 009	3.1	3
	Phase total				24

WBS 4 — Environment & Infrastructure (weeks 3-5 · 13 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
4.1	Repositories & CI/CD pipeline	Repos, build/test/deploy pipeline	NFR-011	2.4	3
4.2	Dev / test environments	Containerized dev and test environments	—	4.1	4
4.3	Provision data & messaging	PostgreSQL, Redis, event bus instances	NFR-004, 008	3.3, 4.2	3
4.4	Logging, metrics & alerting	Dashboards, alert rules, health checks	NFR-004	4.2	3
	Phase total				13

WBS 5 — Backend Development (weeks 5-10 · 61 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
5.1	API gateway, authN / authZ	Gateway, JWT auth, RBAC enforcement	NFR-002, 006	3.6, 4.2	6
5.2	Trip & Member Service	Trip and membership APIs, zero-balance check	FR-004	3.2, 4.3	5
5.3	Expense Service	Expense APIs with custom split validation	FR-002, 007, NFR-005	3.2, 4.3	7
5.4	Currency Conversion Service & rate cache	Conversion logic, Redis rate cache	FR-003, 009	3.2, 4.3	6
5.5	Exchange-Rate Adapter	Adapter with timeout, retry, circuit breaker	NFR-007	5.4	4
5.6	Balance Service	Net balance aggregation and cached view	FR-005, 008, NFR-003	5.3	5
5.7	Settlement Engine	Min-flow engine, precondition checks	FR-001, 013, NFR-001, 005	3.4, 5.6	10
5.8	Payment Tracking Service	Payment confirmation, edge status, settled state	FR-006, 012	5.7	5
5.9	Notification Service & adapter	Event consumers, email / push / in-app delivery	FR-011, 010	4.3, 3.2	6
5.10	Audit log	Append-only audit store and writers	NFR-009	4.3	3
5.11	Rate-refresh & re-settlement job	Scheduled job, RateUpdated handling	FR-010	5.5, 5.7, 5.9	4
	Phase total				61

WBS 6 — Frontend Development (weeks 6-12 · 34 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
6.1	App shell, navigation, sign-in	Responsive app skeleton, auth screens	NFR-010	3.5, 5.1	5
6.2	Manage Travelers screen	Add / remove traveler UI	FR-004	6.1, 5.2	3
6.3	Add Expense & custom split UI	Expense form with amount / % split, currency	FR-002, 003	6.1, 5.3	6
6.4	View Balance screen	Real-time balance view	FR-005	6.1, 5.6	3
6.5	Settle Up & plan review screen	Expense list, balances, settlement plan	FR-007	6.1, 5.7	6
6.6	Confirm payment & settled status UI	Mark paid flow, trip settled state	FR-006, 012	6.5, 5.8	4
6.7	Estimated-rate banner & notifications UI	Warning banner, notification list	FR-009, 010, 011	6.5, 5.9	3
6.8	Responsive & accessibility hardening	WCAG 2.1 AA fixes, cross-browser pass	NFR-010	6.2-6.7	4
	Phase total				34

WBS 7 — Integration (weeks 10-12 · 11 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
7.1	Service-to-service & event flows	End-to-end event flows working	FR-010, 011	5.2-5.11	5
7.2	Live exchange-rate provider	Production rate-provider integration	FR-003, 009	5.5	3
7.3	Email / push provider	Production messaging integration	FR-011	5.9	3
	Phase total				11

WBS 8 — Testing & Quality Assurance (weeks 9-15 · 50 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
8.1	Test plan & test data	Test plan, seed and scenario data	All	2.4	3
8.2	Unit tests (≥ 80% coverage)	Unit suites, coverage gate in CI	NFR-011	5.x	12
8.3	API & integration tests	Automated integration suite	FR-001-013	7.1	8
8.4	Property-based settlement tests	10,000-scenario zero-residual test	FR-001, 008, NFR-005	5.7	4
8.5	Performance & load tests	Benchmarks: 100 ms engine, 2 s views, scale	NFR-001, 003, 008	7.1	5
8.6	Security testing	Encryption scan, RBAC / 403 tests	NFR-002, 006	6.8, 7.1	4
8.7	Fault-injection & DR drill	Outage test, restore drill (RPO / RTO)	NFR-004, 007	7.2	4
8.8	UAT & defect fixing	UAT sign-off, defect log closed	All	8.3-8.7	10
	Phase total				50

WBS 9 — Deployment & Release (weeks 14-16 · 12 person-days)

WBS	Work package	Deliverable	Requirements	Depends on	pd
9.1	Production environment, backup & DR	Prod stack, backups, replica, monitoring	NFR-002, 004	4.3, 8.7	4
9.2	Documentation & runbooks	User guide, operations runbook, API docs	—	8.8	4
9.3	Release, go-live & hypercare	Production release, post-launch support	All	8.8, 9.1	4
	Phase total				12

B.5 Critical path

The longest dependency chain, which determines the 16-week finish, is:

2.4 Requirements sign-off → **3.4** Settlement algorithm design → **5.7** Settlement Engine → **5.8** Payment Tracking → **6.6** Confirm-payment UI → **7.1** Integration → **8.3 - 8.7** Integration, performance, security and resilience tests → **8.8** UAT & defect fixing → **9.3** Release and go-live.

Any slip in the Settlement Engine (5.7) or in UAT defect fixing (8.8) moves the go-live date directly; other tasks have float.

B.6 Risks affecting the plan

ID	Risk	Impact	Mitigation
R1	Exact min-transaction optimization is too slow for 50 travelers (NP-hard).	Misses NFR-001 (100 ms).	Use greedy baseline, bounded optimization; benchmark early (3.4, 8.5).
R2	Exchange-rate provider outage, rate limits or licensing issues.	Delays FR-003 / FR-009.	Cache last good rate; choose provider early; circuit breaker (5.5).
R3	Rounding errors leave non-zero balances.	Breaks FR-001 pass criterion.	Integer minor units; property-based tests (8.4).
R4	Scope growth (expense editing, payment gateway).	Schedule overrun.	Hold scope at SRS baseline; change-control after M1.
R5	Disputes over payment confirmation.	Incorrect “settled” status.	Resolve open issue 2; audit trail (5.10).
R6	Security or privacy findings late in testing.	Rework before go-live.	Threat model in design (3.6); security tests from week 11.
R7	Team capacity shortfall or unavailability.	Slips on critical path.	15% contingency; cross-train backend developers on Settlement Engine.